

Qatar-2b

R-0775

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_240328 · created 2026-07-28T05:35:47+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460397.7708 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.013
Transit depth (Rp/Rs)^2	5.2 +/- 0.58 [%]
Orbital Inclination (inc)	89.73 +/- 1.32
Airmass coefficient 1 (a1)	0.949 +/- 0.037
Airmass coefficient 2 (a2)	0.032 +/- 0.012
Scatter in the residuals of the lightcurve fit is	3.88 %
Best Comparison Star	#1 - [361, 37]
Optimal Method	PSF photometry
Transit Duration (day)	0.0809 +/- 0.0013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.228 ± 0.013 vs 0.16327 (deviation 4.98σ)
Duration fit vs published	0.0809 d vs 0.0767 d (deviation 3.26σ)
Mid-time O-C	-3.3 min
Depth SNR	17.5
Residual scatter	3.88 %

Light curve

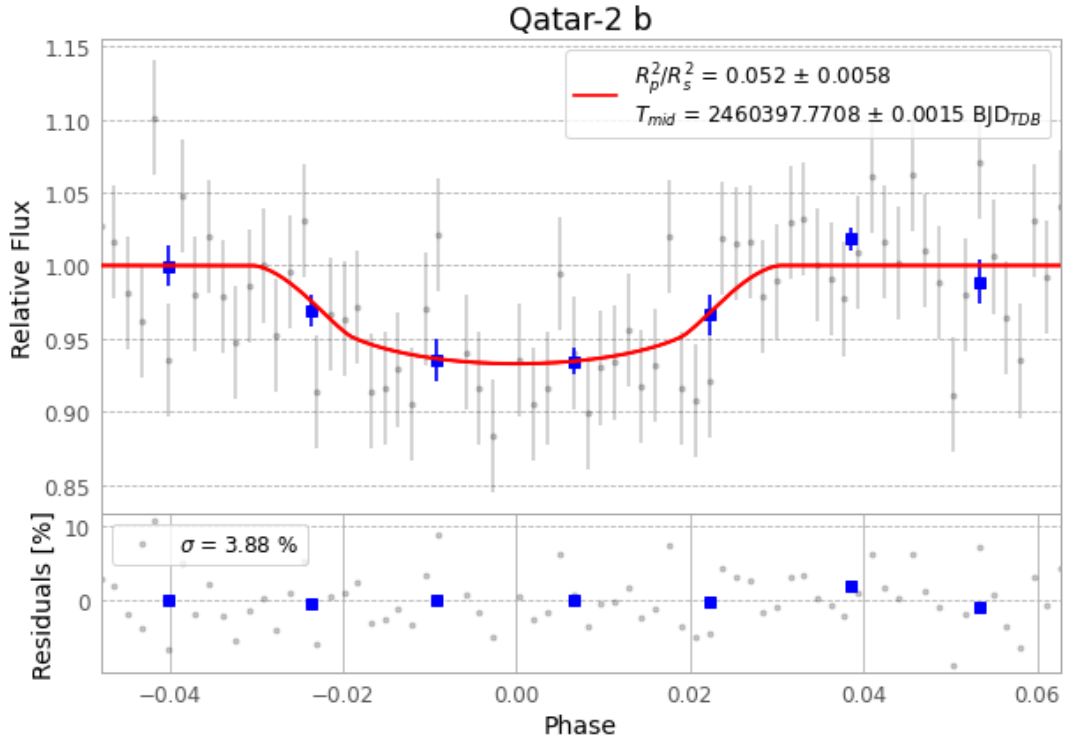


Figure 1 — FinalLightCurve_Qatar-2 b_2024-03-27.png (SHA-256 f5b56346e59d0bdc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [548, 183], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2bf6f5d080fe8265...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

70 FITS frames (46 MB), Qatar-2240328044816.FITS → Qatar-2240328082115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-240328.log (3909f1aed080...), FinalLightCurve_Qatar-2 b_2024-03-27.png (f5b56346e59d...), FinalLightCurve_Qatar-2 b_2024-03-27.csv (773d1403a83a...)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 05:35Z.